Raining Cats & Witches – Project report

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What was done

A cozy, witchy weather app in pixel art style, made with plain HTML, CSS and JavaScript (no frameworks). The user can search for a place or use their GPS location and sees the current weather, an hourly forecast for 24 hours and a 7-day forecast. Weather comes from 4 API providers: **Open-Meteo**, **MET Norway**, **wttr.in** and **OpenWeatherMap**. Every provider file in js/providers/ converts that service's data into the same shape, so the rest of the app works with any source, and if one source fails the others still work.

The background is a moving pixel art scene, like the animated backgrounds in phone weather apps. Its layers drift at different speeds (parallax, faster when it is windy), and the scene is chosen from the weather, the temperature, the time of day (day, golden hour around sunrise/sunset, night) and the moon: for example a sunny beach on hot days, snowy pines on freezing nights and a full moon over the sea on clear full moon nights. Rain and snow are drawn as pixels on a canvas, thunderstorms flash, fog rolls in with a ghost, and bats and a flying witch cross night sky.

The witchy layer adds a **moon card** (phase, % lit, moonrise/moonset, days to full moon and a "will we see it tonight" reading that combines the moon with the night time forecast), a **tarot card of the day** (Tarot API, tap to flip), a **witch's almanac** (sun and moon zodiac signs, countdown to the next Wheel of the Year holiday) and a black cat that gives weather advice.

User manual

- Search a city, or press 📍 for your location.
- °C / °F / K switches units. ☆ adds the city to ★ Favorites.
- Data source picks which weather source the main view shows. Hover or tap the chart to compare sources.
- Tap the tarot card to flip it. Draw again gives new card.
- The background, moon card and the cat’s advice change by themselves with the weather.

Tools used

- HTML, CSS (grid, flexbox, media queries, custom properties, 3D transforms, sprite animations with steps()), JavaScript (fetch, async/await, Canvas, Geolocation, localStorage, Intl).
- **Chart.js 4** for the comparison chart. Google Fonts: Pixelify Sans and Nunito.
- APIs: Open-Meteo (forecast + geocoding), MET Norway, wttr.in, OpenWeatherMap (key), Tarot API, BigDataCloud reverse geocoding.
- Pixel art: CraftPix.net backgrounds, weather icons by bgbw and other free assets (full list with licenses in README.md).
- VS Code, browser developer tools, Claude AI.

Points requested

Feature	Pts	Justification
Search for locations	1	Open-Meteo geocoding, list to choose from when there are several matches.
GPS (Geolocation API)	2	GPS button; coordinates turned into a place name with BigDataCloud.
Two + three providers	3	Open-Meteo, MET Norway, wttr.in and OpenWeatherMap are separate services.
Current weather	1	Temperature, feels like, wind, humidity, precipitation, local date, time.
Next 24 h, hourly	3	Scrollable row with pixel icon, temperature and rain amount.
Next 7 days	3	Day, pixel icon, min / max temperature.
Icons and numbers	3	Pixel icon set for 12 conditions, separate day and night icons from real sunrise/sunset.
Look reflects the weather	2	16 animated scenes chosen by weather, temperature, time of day and moon; rain, snow, fog, lightning.
Two forecasts in a graph	3	24 h temperature lines from all sources in one Chart.js chart.
Favorites menu	2	Add/remove with ☆, saved in localStorage.
°C / °F / K	2	Instant switch without new downloads; remembered.
Well-written PDF report	3	This document
Responsive	4	1 column on phones, 2 on tablets, 3 on desktop
Works in all 4 browsers	3	Application works on Firefox, Safari, Edge and Chrome
Clear directory structure	2	HTML, CSS, JS and assets separated; JS split by topic (see README.md).

Feature	Pts	Justification
Own features	3+	Animated parallax sky; moon card combining the moon with the night forecast; tarot card of the day (5th API); zodiac + Wheel of the Year (hemisphere-aware); weather-reacting cat; switchable data source.
Total requested	**40**	**Maximum for the niche theme and creativity**

Declaration of AI usage

I built this app with huge help of Claude. It was used through as a coding assistant, teacher helping with bottlenecks, helper with my broken logic or suggesting tips how to do feature faster, and also for mathematical formulas and similar heavy logical matters. No other AI tools were used.

Claude wrote either entirely or with huge help all the code where is commented “Challenge” it always comes with explanation for me to learn from it. Claude made most of the precise details in design based on my instructions. Claude rewrote almost all my code for the changing sky background. The problem was that many problems stack on top of each other. Seams were huge bottleneck because a drifting layer must repeat endlessly, but many layers didn’t line up edge-to-edge, so it had to be mirrored which I wouldn’t think of. Scene picking was discussed (and because I wanted many details like changing speed of clouds based on wind speed) later debated with Claude. Rain and snow and fog that were drawn on a canvas were made by Claude. Entirely unexpected challenge of time zones from cities was solved by AI. Helping with APIs own language was made by AI, which saved much time and was precise. AI found a bug: in India (UTC+5:30) MET Norway’s hours were 30 minutes off the local hours, so its whole line vanished. The matching tolerance had to include exactly 30 minutes. Any resizing or mapping with pixel animations was done by AI. Tarot card flipping was made by AI. Problem with tarots cards that numbers didn’t match was solved by AI. Helped with code for different browsers, especially Safari. Persuading me that responsive for phone and laptop/PC isn’t enough and tablet interface should be included. Zodiac logic was figured out by AI. AI pointed out lots of error checkers that I would never think of or didn’t know how to implement correctly.